Dosimetric properties of caesium-137, the probability distribution of decays and the half life of barium-137m

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This experiment compares the radiation of caesium-137 measured directly above its source to be $\dot{D}_{0\mathrm{m}} = (1.16 \pm 0.52) \cdot 10^{-6}\ \frac{\mathrm{Sv}}{\mathrm{h}}$ and in a distance of 0.75 m to be $\dot{D}_{0.75\mathrm{m}} = (0.33 \pm 0.23) \cdot 10^{-6}\ \frac{\mathrm{Sv}}{\mathrm{h}}$. Furthermore, a Geiger-Müller counter was investigated. The threshold voltage was found to be $U_T = (512 \pm 4)$ V. The half life time of barium-137m was measured to be $t_{\frac{1}{2}} = (154.6 \pm 1.2)$ s. By conducting a χ^2-test it was concluded that for both gate times (1 s and 10 s) of the Geiger-Müller counter, the registered emission events caused from an uranium oxide coated tile are Poisson-distributed with a 5 % error probability. Also it was shown that the distribution yielded for a gate time of 10 s can be approximated by a Gaussian distribution.

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1 Introduction

The phenomenon radioactivity has grown to be an essential part of today's technology. Since its discovery at the end of the 19th century it has revolutionised the power generation through fission and healthcare through X-ray imaging and treatments. Furthermore, radioactivity is a key characteristic of the decay of large nuclei. The other important reason for analysing radioactivity is to protect life from it.

Well known examples of its destructive manner are the bombing of Hiroshima and Nagasaki and the nuclear disasters in Chernobyl and Fukoshima.

In the following the element caesium-137 and its decay product barium-137m will be analysed. The dose rate and the activity of caesium-137 were measured, as well as the half-life of barium-137m with consideration of the background radiation and the Null effect. Additionally the probability distribution of the registered γ-quanta per second were examined to either follow the Poission or the Gaußian-distribution.

2 Background radiation

Background radiation consists of cosmic rays like ions or protons and terrestrial sources like potassium, uranium or airborne radon. These radiations were detected by measuring devices and has to be considered for all data. To estimate the background radiation, the dosimeter was placed in a representative location. With the dosimeter in long-term-measurement mode, the radiation was measured at 15-minute intervals throughout the duration of the experiments. This part of the experiment took place, while other measurements with radioactive sources were taken. The Result is

$$\dot{H}_{\mathrm{Background}} = (0.105 \pm 0.017) \cdot 10^{-6}\ \frac{\mathrm{Sv}}{\mathrm{h}}.$$

Therefore, the radiation exposure for four hours is

$$H_{\mathrm{Background}}(4\mathrm{h}) = (0.421 \pm 0.069) \cdot 10^{-6}\ \mathrm{Sv}.$$

If this radiation is constant during one year at this place the radiation dose would be

$$H_{\mathrm{Background}}(1\mathrm{a}) = (0.92 \pm 0.15) \cdot 10^{-3}\ \mathrm{Sv}.$$

Taking into account the errors, this is within the annual radiation limit of 10^{-3} Sv [2] given by law.

3 Dosimetric measurements of caesium-137

3.1 The radioactive isotope caesium-137

All radioactive nuclei decay statistically. The radiation they produce in this process can be classified in three categories. The first one is $\alpha-$radiation. It is a particle radiation of helium-4 nuclei. Due to its mass it can be blocked relatively easily. The second category is $\beta-$radiation. It can be differed in β^- and β^+

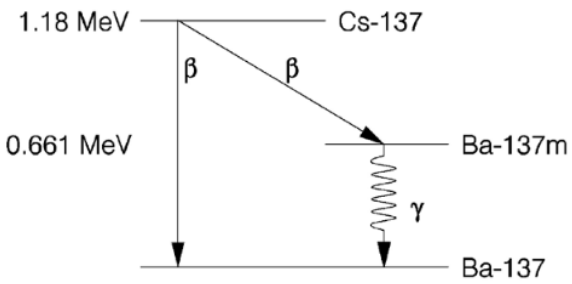

Fig. 1: The diagram shows the decay of caesium-137 into the excited state and ground state of barium-137. Additionally the kind of radiation and the energy of the radiation is shown [3].

decay, which produces an electron or positron respectively. Beta particles have an energy of about 0.5 MeV and are considered less ionizing than alpha particles. The distance they are able to travel depends on the energy they possess. In general, the energy of the beta particle is proportional to the damage they cause in living tissue.

The third radiation is the γ-radiation. Among the three decays described, it is the least ionizing. The rays release energy of a few keV to approximately 8 MeV . The $\gamma-$rays have a high penetration power and can easily pass through the human body. To protect the bone marrow and internal organs a specific kind of shielding made from dense materials, such as lead or concrete, is required.

The examined substance is caesium-137, which is a decay product of the uranium decay chain. It can be detected after an atomic explosion over a great distance due to its ability to travel through air. Caesium-137 has a half-life period of about 30.2 years [3]. It decays with a β-decay partly into a metastable isotope of barium-137. This excited state of barium-137 decays via γ-rays with the energy of 0.661 MeV into its ground state. There is also a small probability of about 5% that caesium-137 decays directly to the ground state of barium-137 by emitting beta radiation. This releases an energy of 1.18 Mev.

3.2 Dose rate of caesium-137

First, the equivalent dose output per unit time of a caesium-137 compound was analyzed. The specimen was stored in a stick of aluminium, which walls were thick enough to shield the emitted β-radiation. To measure the emitted γ-radiation, a NEVA dosimeter was used at a measuring interval of 5 seconds. The dose rate directly above the preparation and at a distance of 75 cm was measured. With the mean value of the dose rate $\dot{D}$ and it's standard error the equivalent dose rate H one would experience over 4 h was calculated, using

$$H = Q \cdot D. \tag{1}$$

Q is a defined factor for each kind of radiation. It varies according the the amount of damage the radiation can cause in the human body. For example the Q for $\gamma-$rays is 1 and for α-rays it would be 20. For $\gamma-$rays, this results in the calculated values

$$\dot{D}_{0\mathrm{m}} = (1.61 \pm 0.52) \cdot 10^{-6} \ \frac{\mathrm{Sv}}{\mathrm{h}}$$
$$H_{0\mathrm{m},4\mathrm{h}} = (6.4 \pm 2.1) \cdot 10^{-6} \ \mathrm{Sv}$$

$$\dot{D}_{0.75\mathrm{m}} = (0.33 \pm 0.23) \cdot 10^{-6} \ \frac{\mathrm{Sv}}{\mathrm{h}}$$
$$H_{0.75\mathrm{m},4\mathrm{h}} = (1.32 \pm 0.92) \cdot 10^{-6} \ \mathrm{Sv}.$$

The background radiation was neglected, as it is covered within the margin of error of the computed dose rate. The maximal dose one is allowed to absorb is defined as 5 mSv per year. To reach this maximal dose a person would need to be exposed directly to the specimen for about (129 ± 42) days with the computed dose rate. If a person is exposed for four hours to this level of radiation they would use up about $(0.129 \pm 0.041)\%$ of their permitted dose about 5 mSv per year.

At a distance of 75 cm, it is not possible to reach the limit value of 5 mSv per year, even at constant exposure. A person who is exposed to this dose rate for four hours uses about $(0.026 \pm 0.018)\%$ of their permitted limit up. This shows that the effect of the emitted γ-radiation decreases significantly with the increasing distance. Nevertheless, it is still advised to avoid radiation whenever it is possible.

3.3 Activity of caesium-137

The activity of caesium-137 will be determined in this part. The activity of a radioactive nuclide is the number of decays in a time interval. It is described in Becquerel (Bq) where 1 Bq equals one decay per second. In general the activity decreases with the time. It is defined as

$$A(t) = \frac{dN(t)}{dt} = -\lambda N(t), \tag{2}$$

with $N(t)$ as the nuclides not yet decayed after the time t and λ as the decaying constant. By solving this first order differential equation the decay law becomes

$$N(t) = N_0 \cdot e^{-\lambda t} = N_0 \cdot e^{\frac{-t}{\tau}}. \tag{3}$$

Here, N_0 is the initial amount of radioactive material and $\tau = \frac{1}{\lambda}$ its mean lifetime. The activity law is defined by these two equations and can be written as

$$A(t) = A_0 \cdot e^{-\lambda t}. \tag{4}$$

Therefore, $A_0 = \lambda N_0$ is the initial activity of the used radioactive material. After a specific time only half of the original nuclides are left, this time is called the half life time

$$t_{\frac{1}{2}} = \frac{\ln(2)}{\lambda}. \tag{5}$$

According to the label the date of production of the caesium-137 specimen was August 1978. The initial activity is noted as $3.70 \cdot 10^6$ Bq. With the given half life time of 30.2 years, the activity of today can be computed to

$$A_{today} = 1.34 \cdot 10^6 \text{ Bq}.$$

4 Geiger-Müller counter

4.1 Characteristics

The counting tube is made of a coaxial outer cathode and an inner anode. It is filled with an ionizable gas. Incoming radiation ionizes the filling gas, due to its high energy. The electric field accelerates the electrons, so that further molecules are ionized. This effect is called avalanche effect and depends on the energy of the ionized particles and the applied voltage. This effect is observed after a threshold. Right after this threshold the Geiger-Müller counter is not able to detect all the created ions. Only ions with greater energy are detected. The reason therefore is, that the electric field inside the coaxial cable is depending on the voltage. With a voltage close to the threshold voltage only high energetic ions are moved. By increasing the voltage, more and more ions are detected. At a certain voltage a relatively stable plateau of pulses is reached. At this voltage all ions are detected.

To investigate the threshold voltage, the Geiger-Müller counter was operated in the range from 500V to 600V. The step size of the digital counter voltage was set to 4V. Therefore the error of the voltage is set to 4V. As shown in fig. 2 the threshold voltage of the counter is

$$U_T = (512 \pm 4) \text{ V}.$$

In the range from 520V to 600V the slope of $m_P = (4.3 \pm 2.3) \cdot 10^{-3}$ Vs indicates a plateau in this range. The chosen operating voltage was $U_W = (560 \pm 4)$ V. This value was chosen, because it is located roughly in the centre of our measured plateau.

4.2 Null-Effect

The natural radiation exposure from cosmic rays, uranium rocks and radon in the air causes a background noise. To determine only the radiation of the considered source, this effect had to be determined. The Geiger-Müller counter detected the decays per second to be

$$N_B = (0.56 \pm 0.24) \ \frac{1}{\text{s}}.$$

For this measurement, all sources were removed if possible and the opening of the counting tube was turned towards the window. Furthermore the Null-Effect in the lead box was measured to be

$$N_L = (0.29 \pm 0.18) \ \frac{1}{\text{s}}$$

For measurements in air and in the lead box, the background radiation is taken into account if necessary.

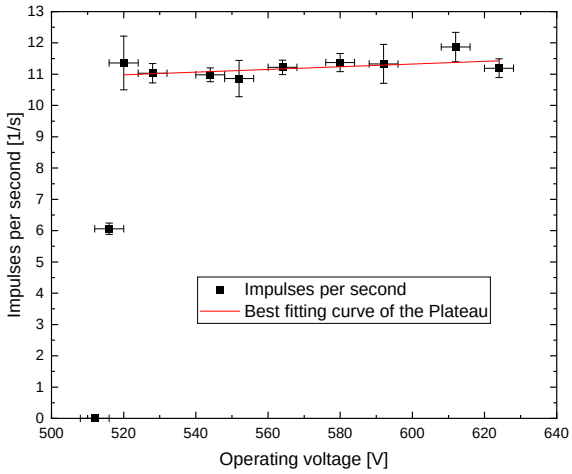

Fig. 2: Shown is the detected rate of pulses with varying voltage in the counter tube. To evaluate the quality of the plateau, a straight line was adjusted in this range. The slope is $m_P = (4.3 \pm 2.3) \cdot 10^{-3}$ Vs. The intersection with the y-axis is at $m_P = (8.7 \pm 1.2) \cdot 10^{-3} \ \frac{1}{\text{s}}$.

4.3 Responsiveness

In this part of the experiment the responsiveness of the Geiger-Müller counter was measured. The responsiveness is determined by the detecting surface of the device and the distance to the radioactive material. The caesium-137 was placed at a distance of $d = (15.00 \pm 0.50)$ cm from the tube. With the computed activity of the caesium-137 from chapter 3 the expected activity is $a = (11.00 \pm 0.50)$ cm^2. Therefore the expected activity is

$$A_{ex} = \frac{a}{4\pi d^2} A = (5.21 \pm 0.42) \cdot 10^3 \text{ Bq}. \tag{6}$$

$$A_{det} = (11.10 \pm 0.77) \text{ Bq}$$

is the detected activity, resulting in a sensitivity of

$$Q = \frac{A_{det}}{A_{ex}} = (0.213 \pm 0.023) \ \% \tag{7}$$

Limitations of the measurement are given by the small opening of the lead cube. Therefore the actual activity of the sample could be bigger.

5 Half-life period of barium-137m

In this experiment the half-life period of the metastable state of barium-137, also labeled as barium-137m was measured. To extract the barium-137m from the caesium-137 compound, the supervisor, Dr. Kießling, injected a hydrochloric acid into the compound with a syringe. The acid flushed the barium-137m out of the caesium-137 compound and thus isolated it for the following examination. The substance was placed in a test tube and the connected LEYBOLD digital counter measured the decays with a gate time of 10 seconds.

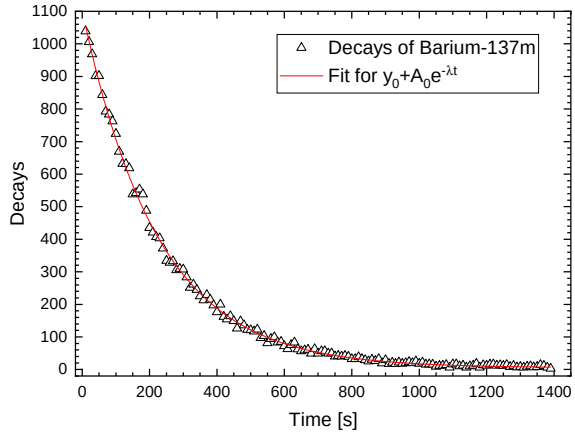

Fig. 3: Shown are the counted decays of barium-137m over a twenty-three minute period. The decay follows the exponential function stated in 8. The decaying constant of $\lambda = (4.4800 \pm 0.0034) \cdot 10^{-3}\ \frac{1}{\mathrm{s}}$, the activity $A_0 = (1099.3 \pm 4.5)$ Bq and $y_0 = (0.50 \pm 0.14)\ \frac{1}{\mathrm{s}}$, which equals the Null-Effect, can be determined.

The number of events was counted roughly over twenty minutes and displayed in figure 3.

It can be seen, that the decay follows the expected exponential function of

$$N(t) = y_0 + A_0 \cdot e^{-\lambda t}. \tag{8}$$

Moreover the decaying constant was evaluated to be

$$\lambda = (4.4800 \pm 0.0034) \cdot 10^{-3}\ \frac{1}{\mathrm{s}},$$

the activity to be

$$A_0 = (1099.3 \pm 4.5)\ \mathrm{Bq}$$

and

$$y_0 = (0.50 \pm 0.14)\ \frac{1}{\mathrm{s}},$$

which equals the Null-Effect. This was already measured in chapter 2. Both values are within the margin of error. The graphically determined value of the Null-Effect is more accurate than the one calculated in chapter 2. With equation 5 the half life of barium-137m can be computed to

$$t_{\frac{1}{2}} = (154.6 \pm 1.2)\ \mathrm{s}.$$

Thus it can be said that within its margin of error, the half-life does not align with the literature value of $t_{\frac{1}{2}} = 2.552$ min $= 153.12$ sec [1]. The reason for the deviation may be that the Geiger-Müller counter is not ideal. It should be noted that the Null-Effect on the measurements for this experiment was ignored. Since its existence would have had no influence on the half life.

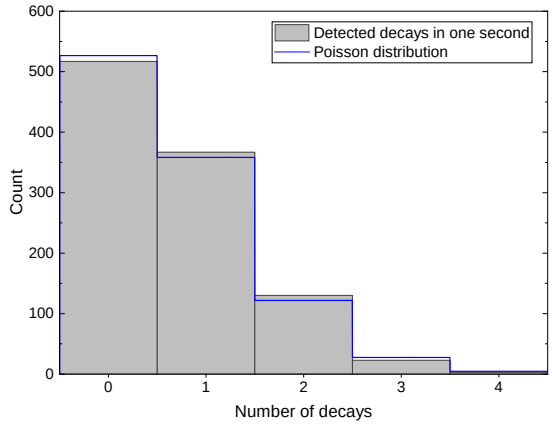

Fig. 4: Probability distribution of detected γ-quanta emitted from a uranium trioxide tile within a time interval of 1s. The expected decays according to a Poisson distribution with mean $\mu_{1\mathrm{s}} = (0.681 \pm 0.025)\ \frac{1}{\mathrm{s}}$ is shown in blue.

6 Probability distribution

To measure the radioactivity of a tile coated with uranium trioxide, a Geiger-Müller counter connected to a personal computer was used. To avoid ambient radiation, the radioactive specimen as well as the Geiger-Müller counter were placed inside a lead chamber. In this setup, the number of decays in a time interval of one and later ten seconds were observed. The decay process is described by a binomial distribution. If, however, the number of trials n goes to infinity while the probability of success p tends to zero, such that $np = \mu_{1\mathrm{s}} = \mathrm{const.}$, the binomial distribution can be approximated by a Poisson distribution

$$P_P(k) = \frac{\mu^k}{k!} e^{-\mu} \tag{9}$$

where k is the number of successes. To verify this probability distribution, a χ^2-test is performed in the following.

6.1 Test for Poisson distribution

To achieve a meaningful result, a number of $n = 1040$ measurements were taken with a gate time of $t = 1$s. The mean value results to $\mu_{1\mathrm{s}} = (0.681 \pm 0.025)\ \frac{1}{\mathrm{s}}$. The result of the measurement can be seen in figure 4. To perform a χ^2-test the 5 occurred classes were combined until their expected value was greater than 5 which leads to 4 considered classes. The sum of the χ_i^2 of each class results in the total of

$$\chi^2_{1\mathrm{s}} = 2.58\ .$$

Since the number of measurements is fixed and the mean had to be calculated from the given data the total degree of freedom for the χ^2-distribution is 2. With a significance level of 5%, the threshold value for the null hypothesis "the present data is Poisson distributed" is $\chi^2 = 5.99$. Therefore, the null hypothesis is accepted.

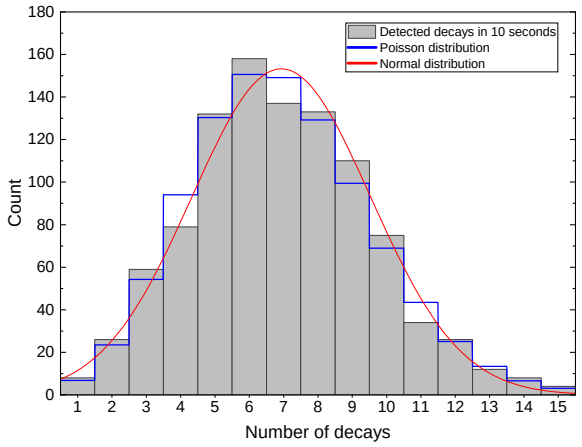

Fig. 5: Probability distribution of detected γ-quanta emitted from a uranium trioxide tile within a time interval of 10 s. The expected decays according to a Gaussian distribution with mean $\mu_{10\mathrm{s}} = (6.931 \pm 0.082)$ and standard deviation $\sigma_{10\mathrm{s}} = 2.6$ is shown in red, the Poisson distribution for the same mean value in blue.

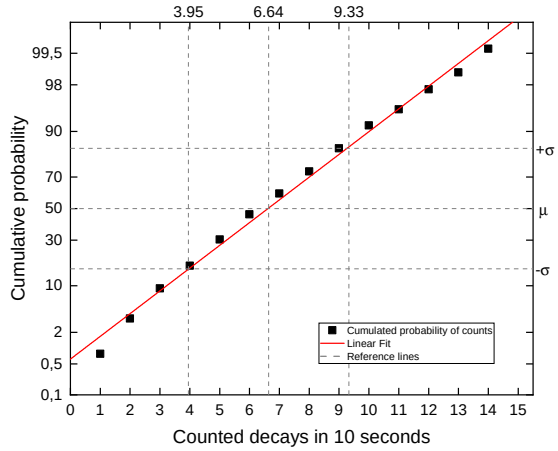

Fig. 6: Cumulative probability of detecting N events with a Geiger-Müller counter with 10 s gate time. To check whether a normal distribution is present, a linear regression has been added; a perfectly normal distributed data set would equate a straight line in this diagram. The experiment was done with an uranium oxide coated tile in a lead chamber.

6.2 Test for Gaussian distribution

Analogously, there were $n = 1001$ measurements taken with a gate time of $t = 10$ s. This results in a mean value increased by a factor of approximately 10. With the measured data, shown in figure 5, the calculated mean value results in $\mu_{10\mathrm{s}} = (6.931 \pm 0.082)$ with the standard deviation $\sigma_{10\mathrm{s}} = 2.61$. Again the 15 classes resulting from the measurements are combined until their expected values are greater than 5. The null hypothesis again is "the present data is Poisson distributed". Here as well, the degrees of freedom are being reduced to a total of 12, since the number of measurements is fixed, and the mean had to be calculated from the given data. for a significance level of 5% the corresponding threshold value is $\chi^2 = 21.03$. Since the given χ^2 is calculated to

$$\chi^2_{10\mathrm{s}} = 8.71,$$

the null hypothesis can be accepted. For a measurement with a mean value of $\mu \geq 9$ or greater the distribution can be described as a normal distribution, defined as

$$P_G(k) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(k-\mu)^2}{2\sigma^2}}. \tag{10}$$

The following should verify that our dataset with mean μ_{10s} can already be approximated as a Gaussian distribution. To investigate this context the cumulative probability is plotted against the number of decays. By using a probability axis the plotted points result in a linear relationship as shown in figure 6. The given correlation coefficient of

$$r = 0.99683$$

shows that the data is well approximated while not exactly described by the Gaussian distribution. The mean value as well as the standard deviation are also determined graphically. The resulting values are

$$\mu_{G,gr.} = (6.640 \pm 0.020)$$
$$\sigma_{gr.} = (2.690 \pm 0.030)$$

with errors resulting from the reading accuracy. With

$$\sigma = \sqrt{\mu} \tag{11}$$

the standard deviation calculated from $\mu_{G,gr.}$ results in $\sigma_{calc.} = (2.5768 \pm 0.0039)$, and is therefore within its error not compatible with the graphically determined value $\sigma_{gr.}$. This also applies to the comparison of the calculated standard deviation $\sigma_{10\mathrm{s}}$ with $\sigma_{gr.}$ and $\sigma_{calc.}$ as well as to the comparison of $\mu_{10\mathrm{s}}$ with $\mu_{G,gr.}$. The fact that those values cannot be agreed within their errors can be justified with the mean value being significantly smaller than 9. Therefore, the Gaussian distribution can only be seen as an approximation but not as an exact description.

7 Summary

In order to exclude the background radiation as a significant source of error, it was determined by long-term measurement. From this measurement the mean annual radiation exposure was calculated to $H_{\mathrm{Background}}(1\mathrm{a}) = (0.92 \pm 0.15) \cdot 10^{-3}$ Sv. With consideration of the errors this is within the annual radiation limit of 1 mSv given by law. A radioactive caesium-137 compound was examined. The Geiger-Müller counter was placed directly above the lead cube the isotope was stored in. With a dosimeter the dose rate caused by γ-emission

was measured to be $\dot{D}_{1\mathrm{m}} = (1.61 \pm 0.52) \cdot 10^{-6}\ \frac{\mathrm{Sv}}{\mathrm{h}}$. In 75 cm distance, the dose rate decreased to only $\dot{D}_{0.75\mathrm{m}} = (0.33 \pm 0.23) \cdot 10^{-6}\ \frac{\mathrm{Sv}}{\mathrm{h}}$. The activity since the date of manufacture has dropped from $3.70 \cdot 10^6$ Bq to $1.34 \cdot 10^6$ Bq. The Geiger-Müller counter itself was examined to reduce possible systematic errors. The the threshold voltage of the counter is $U_T = (512 \pm 4)$ V. The operating voltage was chosen to be $U_W = (560 \pm 4)$ V. This is within the saturation range of the counter. Furthermore, the Null-effect was measured to account for the background radiation and possible set offs of the counter. In air it was determined to be $N_B = (0.56 \pm 0.24)\ \frac{1}{\mathrm{s}}$, which is consistent with the background radiation. Additionally the Null-effect in the lead box is $N_L = (0.29 \pm 0.18)\ \frac{1}{\mathrm{s}}$. The expected activity of caesium-137 is $A_{ex} = (5.21 \pm 0.42) \cdot 10^3$ Bq. The measured activity is $A_{det} = (11.10 \pm 0.77)$ Bq. Both values are for a distance between the source and the counter of $d = (15.0 \pm 0.5)$ cm. Therefore the sensitivity of the Geiger-Müller counter is $Q = (0.213 \pm 0.023)$ %. In order to be able to examine the decay product of the caesium, it was dissolved from the preparation with a hydrochloric solution. The half life time was determined to be $t_{\frac{1}{2}} = (154.6 \pm 1.2)$ s. A chi-square test was performed to check whether the probability distribution for 1s gate time and for 10s gate time is Poisson distributed. With a significance level of 5%, the Poisson distribution can be assumed for both measurements. The mean values of these distributions are $\mu_{1\mathrm{s}} = (0.681 \pm 0.025)$ and $\mu_{10\mathrm{s}} = (6.931 \pm 0.082)$. The measurement for the 10s gate time can be approximated to a Gaussian distribution. By a linearization of the distribution, a correlation coefficient of $r = 0.99683$ can be determined. Therefore the Approximation is reliable.

References

[1] Half-life time of barium-137m, National Nuclear Data Center, last visited on 5th December 2021, https://www.nndc.bnl.gov/nudat2/chartNuc.jsp

[2] Begriffsbestimmung Strahlenschutz, Universität Würzburg, Grundparktikum C2, Versuch 46, last visited on 6th December 2021

[3] Praktikumsanleitung, Versuch C46 Dosimetrie, Halbwertszeit, Abstandsgesetz, last visited 6th December 2021